Optical network simulation software for physical layer design and development

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The development and application of software for modelling the physical layer performance of optical networks is described. The graphical, icon-based user interface allows networks to be constructed and simulated with clarity. Various performance parameters can then be obtained such as power budgets, signal-to-noise ratios, bit error rates and network wavelength response.

1. Introduction

The physical layer design process of optical networks is becoming more and more complicated as we move to amplified, high bit rate, wavelength multiplexed lightwave systems. Not long ago, many optical systems could be designed with a quick power budget calculation and perhaps a simple dispersion estimate. Nowadays, system designs often include passive split amplifiers and some form of WDM (wavelength division multiplexing) which means that the simple calculations no longer apply. The effects of amplifier noise and wavelength response, optical crosstalk (both coherent and incoherent), reflections and nonlinear effects such as four-wave mixing and SRS (stimulated Raman scattering) must now all be considered. Future networks will additionally be optical end-to-end, requiring wavelength agility, routeing and conversion processes.

Many telecommunications companies, including BT, have their own in-house software for modelling optical fibre links. These tools tend to be either power budget based [1, 2] or are designed for detailed bit error rate (BER) prediction of long-haul systems [3, 4], e.g. submarine applications. For detailed BER prediction, the usual simulation method involves the sampling and transmission of a complex signal waveform through the system, using Fourier transform techniques [5]. This is a very powerful method as dispersion, nonlinearities and polarisation properties can all be accounted for in the simulation. The main disadvantage with the technique is that simulation times can be long (minutes to hours), especially for WDM signals where the number of sampling points must be high to maintain frequency resolution.

A simulation methodology has been developed which improves upon this basic technique [6, 7]. This new modelling framework allows detailed BER prediction as described above, but is also versatile enough to allow quick power budget calculations, production of wavelength responses (from 1300 nm to 1600 nm for example) and fast enough to allow detailed design graphs to be obtained for amplified PON (passive optical network) systems [8]. The aim of this paper is to describe the thinking behind the software and to give examples of its use in the design process.

2. Requirements

At the outset, it was decided to make the software ‘easy to use’ in the hands of a skilled person. This required a simple graphical user interface so that the network design could be visualised using icons to represent the component models. In addition, the software needed to be flexible to enable the system design to be explored in various ways. For example, once a network is constructed it should be possible, with little effort, to switch between various modes of simulation, such as:

- an interactive mode where one can ‘play’ with the component parameters while the system is running — this gives instant feedback on the performance variations in terms of numbers and real-time graphical output (this mode is very useful at the initial stages of a design where one is exploring the boundary operating conditions and ball-park operating parameters, and also for interactive demonstration sessions),

- a detailed design mode where it is possible to automatically vary component parameters to produce design graphs — an example might be a contour plot of 10^{-9} BER as a function of splitter loss and fibre length,
- a statistical design mode where the system can be run for a large number of iterations to collect statistical performance data,
- an optimisation mode where parameters can be automatically adjusted to achieve a desired result, e.g. the power of WDM signals input to a fibre amplifier can be automatically adjusted to give equal signal-to-noise ratio for each wavelength at the amplifier output.

3. Modelling framework and methodology

Rather than start from scratch and design our own user interface and modelling framework it was decided to adopt and develop commercial simulation software called SPW from the Alta Group of Cadence Design Systems (formerly Comdisco Systems) [9]. This software is used in many applications areas such as mobile radio design, satellite systems and image processing and has extensive collections of library models for these applications. The capabilities of this simulator have been extended by developing a methodology for modelling optical networks together with a substantial collection of optical component models.

Simulation of optical networks is not entirely straightforward due to the very high bandwidths involved and the need to keep track of broadband amplifier noise and polarisation characteristics. For these reasons every signal on the network is transmitted separately and is only combined with others if necessary (on propagation through a nonlinear element, for example). This approach means that wideband ASE noise and multiple signals can be handled efficiently without the need, in general, to time sample at the Nyquist frequency of the combined multiplex.

Consider, for example, a 1550 nm DFB source modulated at 300 Mbit/s. If there is no laser chirp, the highest information frequency is 150 MHz which leads to a Nyquist sampling rate of 300 MHz. To simulate 128 bits will therefore require 128 time samples. If the source chirps by 0.2 nm, the highest information frequency rises to 25 GHz giving a Nyquist rate of 50 GHz. To simulate 128 bits will now require $\approx 21\,500$ time samples. If a wavelength multiplexed network and/or a system transmitting amplifier noise covering a bandwidth of 60 nm is now considered, the Nyquist rate becomes 7.5 THz. To simulate the 128 bits will now require at least 3.2 million time samples! Simulation soon becomes very inefficient and time consuming. The useful compromise then is to transmit different signals (and noise) separately. Sampling rates can then be tailored to suit each individual signal which leads to much more efficient processing.

4. Optical component models

Many optical component models and utilities have now been written, such as DFB and FP lasers, white-light and spot wavelength sources, single-mode fibre, optical filters, fibre amplifiers (uni and bi-directional), attenuators, reflectors, isolators, couplers, multiplexers, receivers, switches, polarisers, power meters and spectrum analysers. A more complete listing of the models is given in the Appendix. It should be noted that many of these models have the capability of interfacing with those contained in the standard SPW libraries. This feature enhances the application of the software and allows complete integrated designs to be produced bridging both the electrical and optical domains.

In use, the optical component models are selected from the library, placed on the screen and wired together to form the network to be modelled. Each component model has associated with it a parameter screen where the value of the physical model parameters can be set and varied, for example, fibre length or filter bandwidth. Both mean and standard deviation values are specified for every parameter to enable statistical simulation when required. Each parameter value can be specified as a constant or variable which will change in some manner during the simulation — for example, interactively, statistically, read from a list or file, or ramping between given values for graph plotting. The number of simulation iterations between changes of parameters can be controlled so that multidimensional graphs and contour plots can easily be produced.

5. Modelling example — WDM system with gain-clamped, gain-flattened amplifiers

An example WDM system is shown in Fig 1. This system was constructed to simulate one option for the downstream path of a wavelength multiplexed network for the European ACTS (Advanced Communications Technologies and Services) collaborative project, WOTAN (Wavelength-agile Optical Transport and Access Network). The basic component characteristics are given below.

- **Transmitter —** 16 wavelengths ranging from 1538.2 nm to 1562.2 nm with 1.6 nm spacing. The output power of each source is 0 dBm with an extinction ratio of 10 dB. Each source is assumed to be modulated with a PRBS (pseudo-random bit sequence) at 1.4 Gbit/s.
- **Loss A, Loss B and Loss C —** for simplicity, the passive losses in the system are lumped together between the active amplifiers. These losses represent

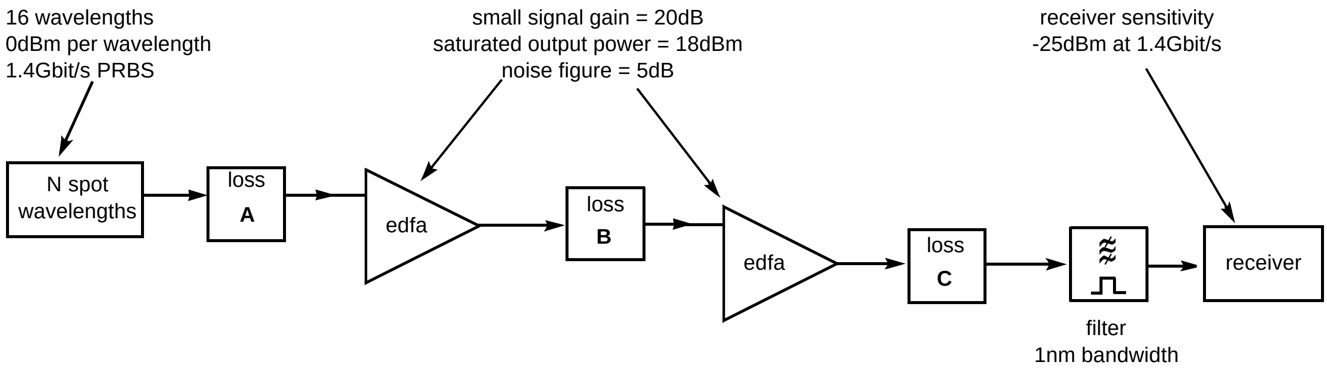

Fig 1 Example WDM system.

fibre loss, splitter loss, splice loss, maintenance budget, operating margin, etc, and are all assumed to be wavelength independent.

- **Erbium-doped fibre amplifiers (EDFAs) —** the fibre amplifiers specified for use in the system are gain controlled and wavelength flattened with a small signal gain of 20 dB, noise figure of 5 dB and a saturated output power of +18 dBm. The amplifier models are based on an average power analysis technique [10] which accounts for both forward and reverse propagating ASE (amplified spontaneous emission) noise. The code describing the basic bi-directional EDFA model is modified to include gain clamping and wavelength flattening. Gain control is effected by reflecting a small quantity of ASE at a defined wavelength back into the cavity [11]. This causes lasing and hence a clamping of the population inversion which defines the gain at all wavelengths. Gain flattening is achieved by placing a filter within the amplifier having a loss approximately equal to the inverse of the gain spectrum. An important characteristic of this comprehensive EDFA model is the small calculation time of approximately half a second (on a SUN SPARCstation 10) for arbitrary input power distribution.

- **Optical filter —** the optical filter is used to select the required wavelength and is initially modelled as a top-hat profile with a bandwidth of 1 nm. The in-band insertion loss is set equal to 0 dB, as loss associated with this filter is assumed to be accounted for in Loss C. The out-of-band insertion loss is set high, at 60 dB, as crosstalk effects are considered separately (see section 5.4).

- **Receiver —** the receiver model estimates the electrical SNR and BER from the input signal and ASE noise. The model takes into account shot noise due to the signal and ASE, signal-spontaneous and spontaneous-spontaneous beat noise and receiver noise. The receiver sensitivity is set to -25 dBm for 10^{-9} BER at a bandwidth of 1.4 Gbit/s. The maximum power allowed on the receiver is -10 dBm.

5.1 Interactive simulation

In order to visualise the performance of the system and to obtain a feel for the operating behaviour it can sometimes be useful to simulate the system in an interactive mode.

Figure 2 is a screen snapshot taken with the system set to interactively display the spectrum before the receiver. The *x*-axis of the spectrum is the wavelength (nm) and the *y*-axis is the power (dBm) in a bandwidth of 0.3 nm. The receiver is set to receive the 1538.2 nm wavelength by the optical filter. The equal level of the other wavelengths confirms the correct operation of the gain-flattened amplifier. The dip in the spectrum around 1532 nm is caused by a filter placed inside the amplifiers to remove the lasing control wavelength.

This mode of operation therefore allows the network to be explored in an interactive manner to confirm correct system operation and to determine the ‘ball-park’ operating limits. This mode is invaluable for use within design meetings and demonstration sessions.

5.2 Detailed power budget design

Once the regions of interest are obtained from the interactive mode, detailed design graphs for the system can be obtained. Loss A and Loss C are automatically varied, for various values of Loss B, while recording BER and the input power to the receiver for every combination. Contour plots of the 10^{-9} BER, -25 dBm received power and -10 dBm received power are then produced. The graph obtained when Loss B = 10 dB is shown in Fig 3. The solid line is the 10^{-9} BER contour. This contour deviates from the -25 dBm received power contour (the receiver sensitivity value) because of the finite extinction ratio and the effects of amplifier ASE. The dotted line represents the maximum power allowed on the receiver. The shaded region thus shows the allowed operating space of the system; the area above and to the right of the shading is where the BER is greater than 10^{-9}, the area below the shading is where the

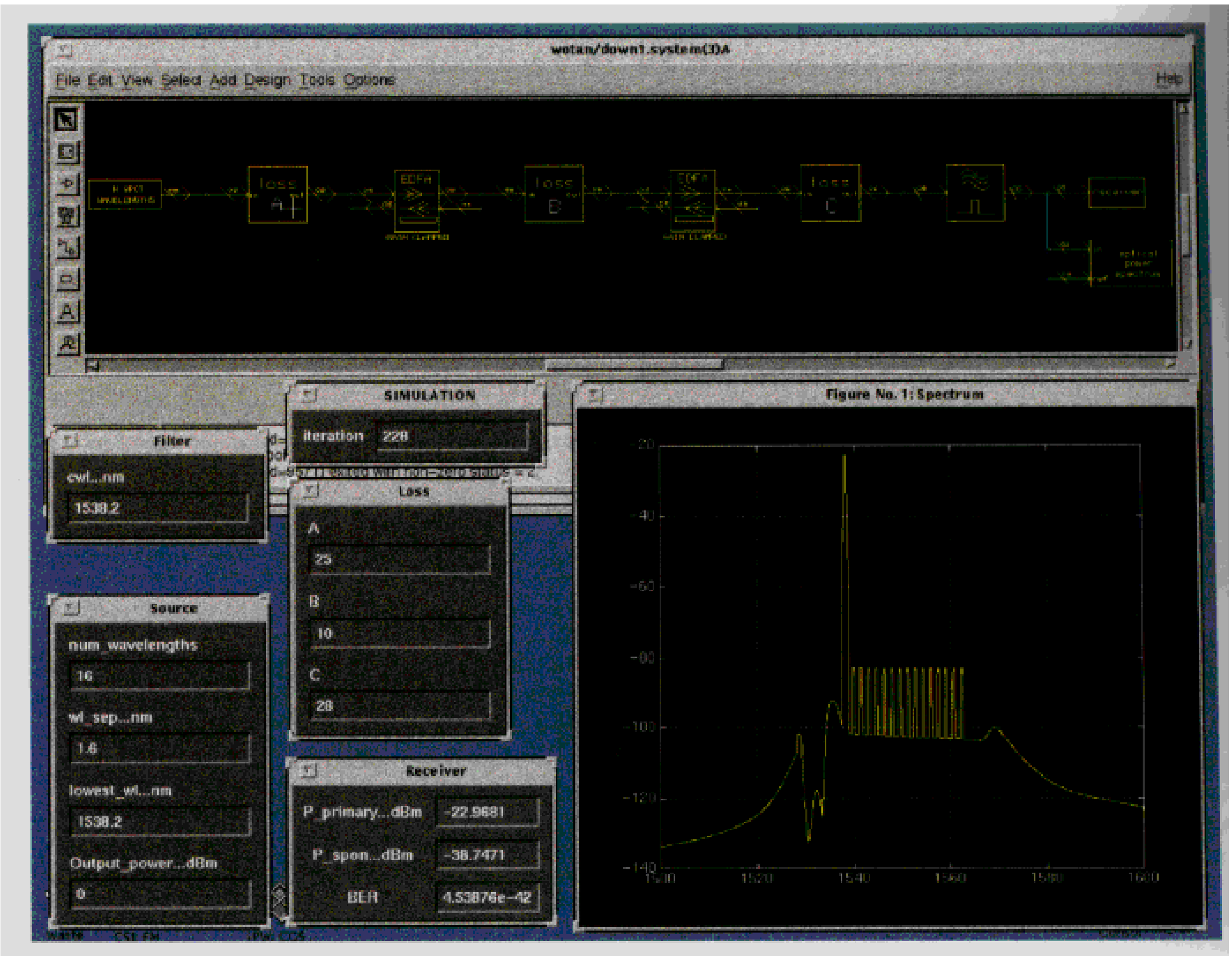

Fig 2 Screen snapshot showing interactive simulation.

total input power to the receiver is greater than its maximum allowed value of –10 dBm, and the area to the left of the shading is where the amplifiers are saturated and are no longer gain controlled or wavelength flat. For this system, it is desirable to maximise the number of customer terminals per amplifier, i.e. Loss C must be as large as possible. The preferred operating area is therefore the upper left corner of the shaded region where Loss C is around 28 dB and Loss A is 25 dB. It should be noted that this is a small operating area — a few dB changes in Loss A will either increase the BER above 10^{-9} or take the amplifier out of the gain-controlled/gain-flattened regime. It is also worth noting that a 3.5 dB decrease in Loss C causes the number of allowed customers to drop by approximately one half.

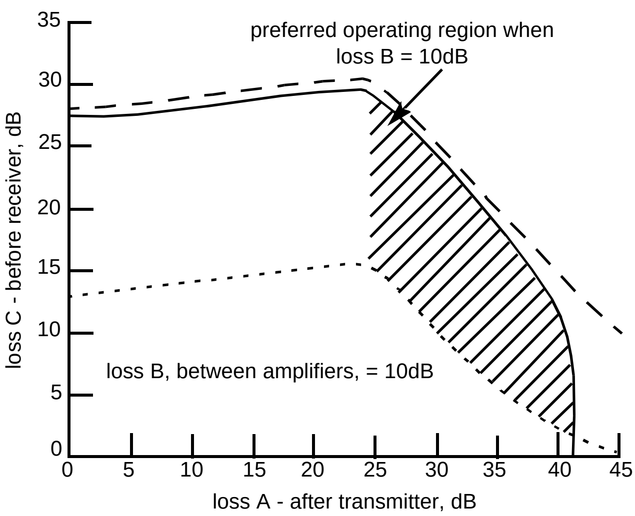

Fig 3 Example design graph which can be obtained by simulating the system shown in Fig 1. The shaded area is the allowed operating region of the system bounded by the 10^{-9} BER (solid-line), the maximum power allowed on the detector (dotted line) and the saturation limit of the amplifiers.

These detailed design graphs are therefore an invaluable aid to system design. They enable one to choose the optimum operating point for the system and to examine the system operating boundaries caused by a number of different factors.

5.3 *Statistical design*

The detailed design has shown that it is important to minimise the operating margin for the system (3.5 dB extra margin means half the number of customers). Every component in the system has variable performance (perhaps specified by a mean and standard deviation value) and it is therefore necessary to design the system with these variations in mind. It may of course be wasteful of capacity to design a system to cope with every component having its worst-case value. An allowed confidence level should therefore be defined for the system of, for example, 95%. This means that, on average, 95% of customer terminals will receive a signal with a BER of less than 10^{-9}. For the remaining 5% of customers, the BER will need to be improved in some way by, for example, increasing receiver sensitivity.

For a straightforward linear system it may be possible to calculate the statistical distribution of losses analytically. In general, however, a Monte-Carlo approach will be necessary where the system is run for a large number of iterations while recording the received BER for each iteration. This is easily done using the simulation software. Every parameter for every component in the system has both a mean and standard deviation value. This allows parameter values to be chosen from normal distributions with ease, although it should be emphasised that the facility exists for values to be chosen from any statistical distribution via specification of its cumulative distribution function in tabular form.

The components which make up Loss C are distribution fibre, split, splices, filter loss, maintenance budget, etc. The allowed values of fibre length and split size need to be calculated for a desired confidence level. This is done by replacing Loss C with a fibre model and a splitter model in addition to a constant loss. The simulation is then run a large number of times for different combinations of split size and fibre length. For every combination the number of times the BER is less than 10^{-9} is recorded. A contour plot showing the desired confidence levels can then be calculated (see Fig 4). This graph shows for example that, for a 10 km distribution fibre length, the split size should be no more than 32 to ensure that 95% of the customers receive a satisfactory signal. The ability to do Monte-Carlo statistical design is important and relies on rapid simulation, as several thousand iterations will normally be required to obtain one data point. To obtain design charts, such as Fig 4, many hundreds of thousands of iterations may be required. These simulations, perhaps in conjunction with suitable cost models, will enable the system design to be optimised for sufficient, but not excessive, operating margins.

5.4 *Crosstalk*

To study the effects of crosstalk due to neighbouring channels, the idealised top-hat filter used previously is replaced with a model where the filter response can be selected to represent either a grating, interference or Fabry-

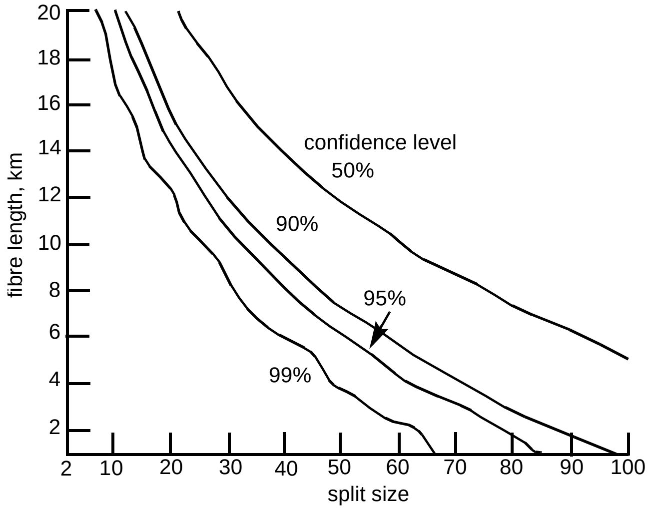

Fig 4 Results of statistical simulation showing 99%, 95%, 90% and 50% confidence levels as a function of split size and fibre length.

Perot device. The interference filter is assumed to be of a three-stage double half-wave design and the Fabry-Perot filter is assumed to have a finesse of 100. With Loss A and B equal to 25 dB and 10 dB respectively, the filter bandwidth and Loss C are varied over a large number of combinations while recording the BER. A contour plot of the 10^{-9} BER is then drawn. Figure 5 shows this graph in the form of a receiver power penalty. To ensure that the power penalty is less than 0.5 dB, the interference, grating and Fabry-Perot filters must have 3 dB bandwidths of less than 2 nm, 1.6 nm and 0.8 nm respectively. The narrow bandwidth of the Fabry-Perot filter is required because of its relatively poor selectivity.

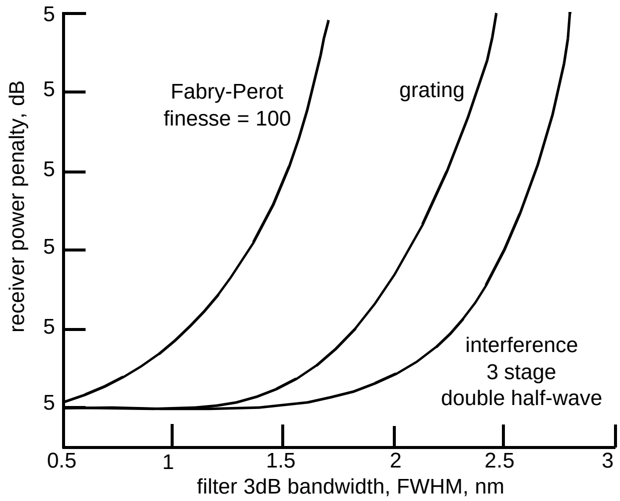

Fig 5 Receiver power penalty for the 16-channel WDM system as a function of filter bandwidth for three filter types. The filter centre wavelength is set in the middle of the 1.6 nm spaced multiplex at 1549.4 nm.

The technique of transmitting signals seperately allows crosstalk effects to be calculated even for widely separated wavelengths. There is no calculation penalty associated with the huge number of time samples that would be required if the signals were combined into a single waveform. Crosstalk between 1300 nm and 1550 nm signals are calculated as easily as if the signals were separated by 1 nm.

6. Future activities

An ongoing activity is experimental validation and development of the models and simulation environment. Collaborative projects currently exist with two UK universities for independent validation and contributions to development. To allow a wider range of people to access the models, such as network planners, it is hoped to provide versions of the software which will run on a PC and possibly a MAC platform. As SPW ™ is not available for these platforms, the search for a different host is currently in progress.

7. Conclusions

A methodology and library of optical component models has been developed for simulating the physical layer performance of optical fibre networks. The software environment allows network models to be constructed easily and efficiently using a graphical, icon-based user interface. The networks can then be simulated in a variety of ways to obtain performance parameters such as power budgets, spectral response, signal-to-noise ratios and bit error rates, etc. The simulation technique can be easily switched to allow interactive demonstration, statistical ‘Monte-Carlo’ simulation or the calculation of detailed performance charts. Multidimensional and contour plots of complex systems can easily be obtained due to the efficient method of simulation. It should be noted that this is possible even when the system includes multiple optical fibre amplifiers where gain, ASE and saturation power effects are all taken into account.

The software currently runs on a SUN workstation as an add-on library to the commercial simulator, SPW ™. To enable the software to be used on a more universal basis by network planners, for example, work is now under way to produce a version of the optical library which will run on a PC.

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Appendix

The following list contains details of the currently available models.

Sources
- Bit stream generator
- Pseudo-random sequence generator
- Laser driver (bit stream to current)
- DFB laser
- FP laser
- Spot wavelength (rectangular, gaussian or lorentzian lineshape)
- Spot wavelength array
- Whitelight source

Fibre
- Linear single-mode with spectral attenuation and dispersion.

Filters
- Rectangular, Fabry-Perot, interference, grating and arbitrary profile

Amplifiers
- Uni-directional EDFA
- Bi-directional EDFA
- Gain-clamped/gain-flattened EDFA
- Power multiplier

Attenuation
- Arbitrary loss, wavelength dependent or constant
- Reflection
- Optical isolator
- Linear polariser

Couplers
- 2×2 coupler with spectral dependence
- $N \times N$ coupler, arbitrary N

Multiplexers
- 2×1 ideal multiplexer
- 4×1 ideal multiplexer
- 4×1 real multiplexer with input filtering and loss
- Add/drop multiplexer

Receivers
- Analytic receiver, assumes PRBS and accounts for beat noise, crosstalk and receiver noise
- Analytic receiver for sub-carrier multiplexed systems
- Dynamic receiver for displaying received signal waveforms and eye-diagrams

Instruments
- Spectrum analyser
- Optical power meter
- Amplifier gain, power and noise figure calculator

Switches
- 1×2, 2×1, 2×2
- 1×8, 8×1

Miscellaneous

Spectrum save and retrieve from file
Matlab interface for interactive plots
One iteration delay for creation of feedback loops
Spectrum propagation delay

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David Mortimore received the BSc degree in applied physics from the City University, London in 1979 and a PhD in electrical engineering from Essex University in 1991. After his first degree he worked in the Laser Systems department of Marconi Avionics Ltd, on advanced lasers and laser systems. He joined BT in 1982 and, after a short time with BT International, he joined the Optical Communications Research Department at BT Laboratories.

His research work involved the fabrication and design of novel fused fibre devices, such as capillary clad monolithic 1 × N couplers, and wavelength flattened couplers for which he received a Gordon Radley award. In 1992 he started a new area of work concerned with the development of tools and techniques for simulating the physical layer performance of optical networks.

He is a Chartered Engineer, Chartered Physicist, Member of the IEE and Member of the IOP.